

Text Files

Writing a Text File

```
int main()
{
    ofstream outFile( "grades.txt", ios::out );

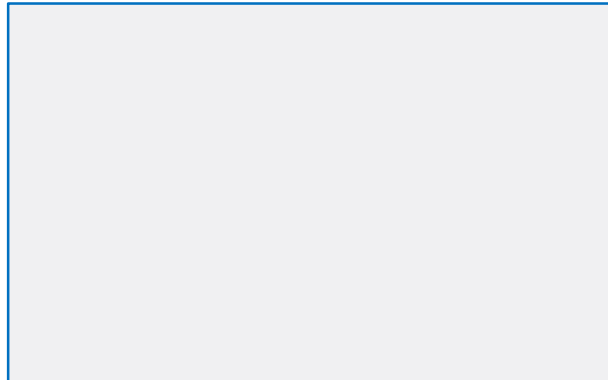
    if( !outFile )
    {
        cout << "File could not be opened\n";
        system( "pause" );
        exit( 1 );
    }

    string id[] = { "1141401", "1141409", "1141424", "1143340" };
    string name[] = { "Tracy", "Wendy", "Sammy", "Melody" };
    int grade[] = { 75, 88, 100, 90 };

    for( int i = 0; i < 4; i++ )
        outFile << id[ i ] << ' ' << name[ i ] << ' ' << grade[ i ]
                << endl;
}
```

Writing a Text File

```
string id[] = { "1141401", "1141409", "1141424", "1143340" };  
string name[] = { "Tracy", "Wendy", "Sammy", "Melody" };  
int grade[] = { 75, 88, 100, 90 };  
for( int i = 0; i < 4; i++ )  
    outFile << id[ i ] << ' ' << name[ i ] << ' ' << grade[ i ] << endl;
```

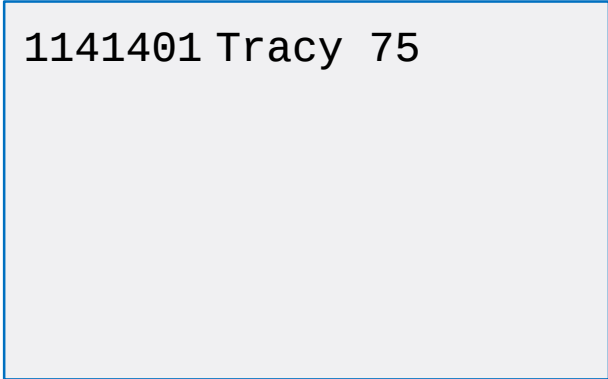


grades.txt

	id	name	grade
0	1141401	Tracy	75
1	1141409	Wendy	88
2	1141424	Sammy	100
3	1143340	Melody	90

Writing a Text File

```
string id[] = { "1141401", "1141409", "1141424", "1143340" };  
string name[] = { "Tracy", "Wendy", "Sammy", "Melody" };  
int grade[] = { 75, 88, 100, 90 };  
for( int i = 0; i < 4; i++ )  
    outFile << id[ i ] << ' ' << name[ i ] << ' ' << grade[ i ] << endl;
```



1141401 Tracy 75

grades.txt

	id	name	grade
0	1141401	Tracy	75
1	1141409	Wendy	88
2	1141424	Sammy	100
3	1143340	Melody	90

Writing a Text File

```
string id[] = { "1141401", "1141409", "1141424", "1143340" };  
string name[] = { "Tracy", "Wendy", "Sammy", "Melody" };  
int grade[] = { 75, 88, 100, 90 };  
for( int i = 0; i < 4; i++ )  
    outFile << id[ i ] << ' ' << name[ i ] << ' ' << grade[ i ] << endl;
```

```
1141401 Tracy 75  
1141409 Wendy 88
```

grades.txt

	id	name	grade
0	1141401	Tracy	75
1	1141409	Wendy	88
2	1141424	Sammy	100
3	1143340	Melody	90

Writing a Text File

```
string id[] = { "1141401", "1141409", "1141424", "1143340" };  
string name[] = { "Tracy", "Wendy", "Sammy", "Melody" };  
int grade[] = { 75, 88, 100, 90 };  
for( int i = 0; i < 4; i++ )  
    outFile << id[ i ] << ' ' << name[ i ] << ' ' << grade[ i ] << endl;
```

```
1141401 Tracy 75  
1141409 Wendy 88  
1141424 Sammy 100
```

grades.txt

	id	name	grade
0	1141401	Tracy	75
1	1141409	Wendy	88
2	1141424	Sammy	100
3	1143340	Melody	90

Writing a Text File

```
string id[] = { "1141401", "1141409", "1141424", "1143340" };  
string name[] = { "Tracy", "Wendy", "Sammy", "Melody" };  
int grade[] = { 75, 88, 100, 90 };  
for( int i = 0; i < 4; i++ )  
    outFile << id[ i ] << ' ' << name[ i ] << ' ' << grade[ i ] << endl;
```

```
1141401 Tracy 75  
1141409 Wendy 88  
1141424 Sammy 100  
1143340 Melody 90
```

grades.txt

	id	name	grade
0	1141401	Tracy	75
1	1141409	Wendy	88
2	1141424	Sammy	100
3	1143340	Melody	90

Writing a Text File

- `!outFile`
 - Returns `true` if `failbit` or `badbit` set.
 - Opened non-existent file for reading, wrong permissions
- `cin >> myVariable`
 - `false` when `failbit` or `badbit` set, otherwise `true`.
 - `failbit` set when end-of-file indicator (*`ctrl-z`*) is entered.
- `while( cin >> myVariable )`
 - Loops until end-of-file indicator (*`ctrl-z`*) is entered.

file stream classes constructors

```
fstream( const char* filename,  
         ios_base::openmode mode = ios_base::in | ios_base::out );
```

```
fstream( const string &filename,  
         ios_base::openmode mode = ios_base::in | ios_base::out );
```

```
ifstream( const char* filename,  
          ios_base::openmode mode = ios_base::in );
```

```
ifstream( const string &filename,  
          ios_base::openmode mode = ios_base::in );
```

```
ofstream( const char* filename,  
          ios_base::openmode mode = ios_base::out );
```

```
ofstream( const string &filename,  
          ios_base::openmode mode = ios_base::out );
```

Parameters

`filename`

A string representing the name of the file to be opened.

Specifics about its format and validity depend on the library implementation and running environment.

`mode`

Flags describing the requested input/output mode for the file.

This is an object of the bitmask member type `openmode` that consists of a combination of the following member constants:

member constant	stands for	access
<code>in *</code>	input	File open for reading: the <i>internal stream buffer</i> supports input operations.
<code>out</code>	output	File open for writing: the <i>internal stream buffer</i> supports output operations.
<code>binary</code>	binary	Operations are performed in binary mode rather than text.
<code>ate</code>	at end	The <i>output position</i> starts at the end of the file.
<code>app</code>	append	All output operations happen at the end of the file, appending to its existing contents.
<code>trunc</code>	truncate	Any contents that existed in the file before it is open are discarded.

These flags can be combined with the bitwise OR operator (`|`).

* `in` is always set for `ifstream` objects (even if not set in argument *mode*).

Note that even though `ifstream` is an input stream, its internal `filebuf` object may be set to also support output operations.

Read One Record

```
int main()
{
    ifstream inFile( "grades.txt", ios::in );

    if( !inFile )
    {
        cout << "File could not be opened" << endl;
        system( "pause" );
        exit( 1 );
    }

    char id[ 8 ];    // student's id
    char name[ 8 ]; // student's name
    int grade;       // student's grade

    inFile >> id >> name >> grade;
    cout << left << setw( 9 ) << id << setw( 8 ) << name
         << right << setw( 5 ) << grade << endl;
}
```

Read All Records

```
int main()
{
    ifstream inFile( "grades.txt", ios::in );

    if( !inFile )
    {
        cout << "File could not be opened" << endl;
        system( "pause" );
        exit( 1 );
    }

    string id;    // student's id
    string name;  // student's name
    int grade;    // student's grade

    cout << left << setw( 9 ) << "Id" << setw( 8 )
         << "Name" << "Grade" << endl;

    while( inFile >> id >> name >> grade )
        cout << left << setw( 9 ) << id << setw( 8 ) << name
             << right << setw( 5 ) << grade << endl;
}
```

```
int main()
{
    ifstream inFile( "grades.txt", ios::in );

    if( !inFile )
    {
        cout << "File could not be opened" << endl;
        system( "pause" );
        exit( 1 );
    }

    string id;    // student's id
    string name;  // student's name
    int grade;    // student's grade

    inFile >> id >> name >> grade;
    while( !inFile.eof() )
    {
        cout << left << setw( 9 ) << id << setw( 8 ) << name
              << right << setw( 5 ) << grade << endl;
        inFile >> id >> name >> grade;
    }
}
```

```
while( inFile >> id >> name >> grade )  
    cout << left << setw( 9 ) << id << setw( 8 ) << name  
        << right << setw( 5 ) << grade << endl;
```

```
1141401 Tracy 75  
1141409 Wendy 88  
1141424 Sammy 100  
1143340 Melody 90
```

grades.txt

id	1141401
name	Tracy
grade	75

Read All Records

Id	Name	Grade
----	------	-------

Output


```
while( inFile >> id >> name >> grade )  
    cout << left << setw( 9 ) << id << setw( 8 ) << name  
        << right << setw( 5 ) << grade << endl;
```

```
1141401 Tracy 75  
1141409 Wendy 88  
1141424 Sammy 100  
1143340 Melody 90
```

grades.txt

id	1141401
name	Tracy
grade	75

Id	Name	Grade
1141401	Tracy	75

Output

Read All Records

```
while( inFile >> id >> name >> grade )  
    cout << left << setw( 9 ) << id << setw( 8 ) << name  
        << right << setw( 5 ) << grade << endl;
```

```
1141401 Tracy 75  
1141409 Wendy 88  
1141424 Sammy 100  
1143340 Melody 90
```

grades.txt

id	1141409
name	Wendy
grade	88

Id	Name	Grade
1141401	Tracy	75

Output

Read All Records

```
while( inFile >> id >> name >> grade )  
    cout << left << setw( 9 ) << id << setw( 8 ) << name  
        << right << setw( 5 ) << grade << endl;
```

```
1141401 Tracy 75  
1141409 Wendy 88  
1141424 Sammy 100  
1143340 Melody 90
```

grades.txt

id	1141409
name	Wendy
grade	88

Id	Name	Grade
1141401	Tracy	75
1141409	Wendy	88

Output

Read All Records

```
while( inFile >> id >> name >> grade )  
    cout << left << setw( 9 ) << id << setw( 8 ) << name  
        << right << setw( 5 ) << grade << endl;
```

```
1141401 Tracy 75  
1141409 Wendy 88  
1141424 Sammy 100  
1143340 Melody 90
```

grades.txt

id	1141424
name	Sammy
grade	100

Id	Name	Grade
1141401	Tracy	75
1141409	Wendy	88

Output

Read All Records

```
while( inFile >> id >> name >> grade )  
    cout << left << setw( 9 ) << id << setw( 8 ) << name  
        << right << setw( 5 ) << grade << endl;
```

```
1141401 Tracy 75  
1141409 Wendy 88  
1141424 Sammy 100  
1143340 Melody 90
```

grades.txt

id	1141424
name	Sammy
grade	100

Id	Name	Grade
1141401	Tracy	75
1141409	Wendy	88
1141424	Sammy	100

Output

Read All Records

```
while( inFile >> id >> name >> grade )
    cout << left << setw( 9 ) << id << setw( 8 ) << name
        << right << setw( 5 ) << grade << endl;
```

```
1141401 Tracy 75
1141409 Wendy 88
1141424 Sammy 100
1143340 Melody 90
```

grades.txt

id	1143340
name	Melody
grade	90

Id	Name	Grade
1141401	Tracy	75
1141409	Wendy	88
1141424	Sammy	100

Output

Read All Records

```
while( inFile >> id >> name >> grade )
    cout << left << setw( 9 ) << id << setw( 8 ) << name
        << right << setw( 5 ) << grade << endl;
```

```
1141401 Tracy 75
1141409 Wendy 88
1141424 Sammy 100
1143340 Melody 90
```

grades.txt

id	1143340
name	Melody
grade	90

Id	Name	Grade
1141401	Tracy	75
1141409	Wendy	88
1141424	Sammy	100
1143340	Melody	90

Output

Read All Records

Open a Text File

```
ifstream inFile( "grades.txt", ios::in );
```

Open an existing file; if the file doesn't exist, fail.

```
ofstream outFile( "grades.txt", ios::out );
```

Create a file; if the file has existed, the data in it will be deleted

Open a Text File

```
ifstream inFile( "grades.txt", ios::in );  
ifstream inFile;  
inFile.open( "grades.txt", ios::in );
```

```
ofstream outFile( "grades.txt", ios::out );  
ofstream outFile;  
outFile.open( "grades.txt", ios::out );
```

File Opening Errors

- Some possible file opening errors are:
 - attempting to open a nonexistent file for reading,
 - attempting to open a file for reading or writing without permission and
 - opening a file for writing when no disk space is available.

File Position

```
int main()
{
    ifstream inFile( "grades.txt", ios::in );

    if( !inFile )
    {
        cout << "File could not be opened" << endl;
        system( "pause" );
        exit( 1 );
    }

    string id;    // student's id
    string name;  // student's name
    int grade;    // student's grade

    cout << left << setw( 9 ) << "Id" << setw( 8 )
         << "Name" << "Grade" << endl;

    while( inFile >> id >> name >> grade )
        cout << left << setw( 9 ) << id << setw( 8 ) << name
             << right << setw( 5 ) << grade << endl;
}
```

```

while( inFile >> id >> name >> grade )
    cout << left << setw( 9 ) << id
        << setw( 8 ) << name
        << right << setw( 5 ) << grade
        << endl;

```

file position

0

grades.txt

```

1141401 Tracy 75
1141409 Wendy 88
1141424 Sammy 100
1143340 Melody 90

```

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7	00100000	32	
8	01010100	84	T
9	01110010	114	r
10	01100001	97	a
11	01100011	99	c
12	01111001	121	y
13	00100000	32	
14	00110111	55	7
15	00110100	53	5
16			

```

while( inFile >> id >> name >> grade )
    cout << left << setw( 9 ) << id
        << setw( 8 ) << name
        << right << setw( 5 ) << grade
        << endl;

```

file position

8

grades.txt

```

1141401 Tracy 75
1141409 Wendy 88
1141424 Sammy 100
1143340 Melody 90

```

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7	00100000	32	
8	01010100	84	T
9	01110010	114	r
10	01100001	97	a
11	01100011	99	c
12	01111001	121	y
13	00100000	32	
14	00110111	55	7
15	00110100	53	5
16			

```

while( inFile >> id >> name >> grade )
    cout << left << setw( 9 ) << id
        << setw( 8 ) << name
        << right << setw( 5 ) << grade
        << endl;

```

file position

14

grades.txt

```

1141401 Tracy 75
1141409 Wendy 88
1141424 Sammy 100
1143340 Melody 90

```

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7	00100000	32	
8	01010100	84	T
9	01110010	114	r
10	01100001	97	a
11	01100011	99	c
12	01111001	121	y
13	00100000	32	
14	00110111	55	7
15	00110100	53	5
16			


```

while( inFile >> id >> name >> grade )
    cout << left << setw( 9 ) << id
        << setw( 8 ) << name
        << right << setw( 5 ) << grade
        << endl;

```

file position

16

grades.txt

```

1141401 Tracy 75
1141409 Wendy 88
1141424 Sammy 100
1143340 Melody 90

```

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7	00100000	32	
8	01010100	84	T
9	01110010	114	r
10	01100001	97	a
11	01100011	99	c
12	01111001	121	y
13	00100000	32	
14	00110111	55	7
15	00110100	53	5
16			

Writing a Text File

```
int main()
{
    ofstream outFile( "grades.txt", ios::out );

    if( !outFile )
    {
        cout << "File could not be opened\n";
        system( "pause" );
        exit( 1 );
    }

    string id[] = { "1141401", "1141409", "1141424", "1143340" };
    string name[] = { "Tracy", "Wendy", "Sammy", "Melody" };
    int grade[] = { 75, 88, 100, 90 };

    for( int i = 0; i < 4; i++ )
        outFile << id[ i ] << ' ' << name[ i ] << ' ' << grade[ i ]
                << endl;
}
```

```
for( int i = 0; i < 4; i++ )
    outFile << id << ' ' << name
            << ' ' << grade << endl;
```

file position

0

grades.txt

id 1141401

name Tracy

grade 75

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7	00100000	32	
8	01010100	84	T
9	01110010	114	r
10	01100001	97	a
11	01100011	99	c
12	01111001	121	y
13	00100000	32	
14	00110111	55	7
15	00110100	53	5
16			

```

for( int i = 0; i < 4; i++ )
    outFile << id << ' ' << name
            << ' ' << grade << endl;

```

file position

7

grades.txt

1141401

id	1141401
name	Tracy
grade	75

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			

```
for( int i = 0; i < 4; i++ )
    outFile << id << ' ' << name
            << ' ' << grade << endl;
```

file position

8

grades.txt

1141401

id	1141401
name	Tracy
grade	75

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7	00100000	32	
8			
9			
10			
11			
12			
13			
14			
15			
16			

```

for( int i = 0; i < 4; i++ )
    outFile << id << ' ' << name
            << ' ' << grade << endl;

```

file position

13

grades.txt

1141401 Tracy

id	1141401
name	Tracy
grade	75

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7	00100000	32	
8	01010100	84	T
9	01110010	114	r
10	01100001	97	a
11	01100011	99	c
12	01111001	121	y
13			
14			
15			
16			

```

for( int i = 0; i < 4; i++ )
    outFile << id << ' ' << name
            << ' ' << grade << endl;

```

file position

14

grades.txt

1141401 Tracy

id	1141401
name	Tracy
grade	75

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7	00100000	32	
8	01010100	84	T
9	01110010	114	r
10	01100001	97	a
11	01100011	99	c
12	01111001	121	y
13	00100000	32	
14			
15			
16			


```

for( int i = 0; i < 4; i++ )
    outFile << id << ' ' << name
            << ' ' << grade << endl;

```

file position

16

grades.txt

1141401 Tracy 75

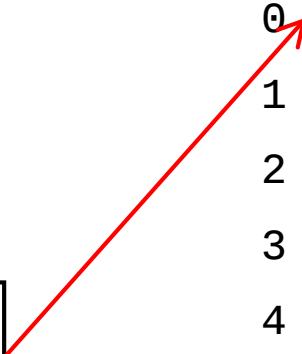
id	1141401
name	Tracy
grade	75

0	00110001	49	1
1	00110001	49	1
2	00110100	52	4
3	00110001	49	1
4	00110100	52	4
5	00110000	48	0
6	00110001	49	1
7	00100000	32	
8	01010100	84	T
9	01110010	114	r
10	01100001	97	a
11	01100011	99	c
12	01111001	121	y
13	00100000	32	
14	00110111	55	7
15	00110100	53	5
16			


```
inFile.seekg( 0, ios::beg );
```

file position

0



0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			

```
inFile.seekg( 2, ios::beg );
```

file position

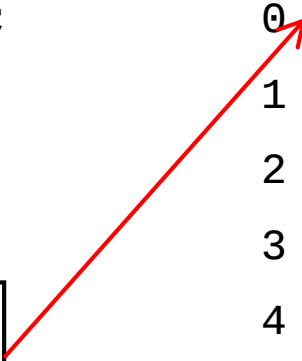
2

0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			

```
inFile.seekg( -1, ios::beg );
```

file position

0



0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			

```
inFile.seekg( -1, ios::beg )
```

sets the failbit and resets the goodbit.

```
inFile.seekg( 0, ios::end );
```

file position

8

0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			

```
inFile.seekg( -1, ios::end );
```

file position

7

0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			

```
inFile.seekg( -3, ios::end );
```

file position

5

0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			


```
inFile.seekg( 1, ios::end );
```

file position

9

0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			

```
inFile.seekg( 2, ios::cur );
```

file position

4

0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			

```
inFile.seekg( 2, ios::cur );
```

file position

6

0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			

```
inFile.seekg( -1, ios::cur );
```

file position

4

0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			

```
inFile.seekg( -1, ios::cur );
```

file position

3

0	01100001	97	a
1	01100010	98	b
2	01100011	99	c
3	01100100	100	d
4	01100101	101	e
5	01100110	102	f
6	01100111	103	g
7	01101000	104	h
8			
9			
10			
11			